

Дефиниция

Нека $f(x) \in F[x]$ и $\alpha \in F$. Тогава α е корен на $f(x)$, ако $f(\alpha) = 0 \Leftrightarrow x - \alpha \mid f(x)$

Дефиниция

Нека $f(x) \in F[x]$. Тогава $\alpha \in F$ е k -кратен корен на $f(x)$ ако: $(x - \alpha)^k \mid f(x)$ и $(x - \alpha)^{k+1} \nmid f(x)$

Дефиниция

Нека $f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n \in F[x]$. Тогава производни на $f(x)$ е

$$f'(x) = a_0 n x^{n-1} + a_1 (n-1) x^{n-2} + \dots + a_{n-1}$$

Твърдение

Нека $f(x) \in F[x]$, тогава $\alpha \in F$ е k -кратен $\Leftrightarrow f(\alpha) = f'(\alpha) = 0$

Доказателство

$$\Rightarrow (x - \alpha)^2 \mid f(x) \Rightarrow f(x) = (x - \alpha)^2 g(x)$$

$$f'(x) = 2(x - \alpha)g(x) + (x - \alpha)^2 g'(x) = (x - \alpha)(2g(x) + (x - \alpha)g'(x)) \Rightarrow f'(\alpha) = f(\alpha) = 0$$

$$\Leftarrow f(\alpha) = 0 \Rightarrow (x - \alpha) \mid f(x) \Rightarrow f(x) = (x - \alpha)h(x) \text{ и } h(x) = (x - \alpha)b(x) + c$$

$$f(x) = (x - \alpha)h(x) = (x - \alpha)((x - \alpha)b(x) + c) = (x - \alpha)^2 b(x) + (x - \alpha)c$$

$$f'(x) = 2(x - \alpha)b(x) + (x - \alpha)^2 b'(x) + c \Rightarrow f'(\alpha) = 0 + 0 + c, \text{ но } f'(\alpha) = 0 \Rightarrow c = 0 \Rightarrow (x - \alpha) \mid h(x) \Rightarrow (x - \alpha)^2 \mid f(x)$$

Принцип за сравняване на коефициентите

Нека F е поле и $f(x), g(x) \in F[x]$, $\deg f \leq n$ и $\deg g \leq n$

Ако $a_1, a_2, \dots, a_{n+1} \in F$ и $a_i = a_j \Leftrightarrow i = j$ и накрая, че $f(a_i) = g(a_i)$, $1 \leq i \leq n+1$, то $f(x) = g(x)$

Доказателство

$$(x - a_i, x - a_j) = 1$$

Нека $h(x) = f(x) - g(x) \Rightarrow \deg h \leq n$

$$h(a_i) = f(a_i) - g(a_i) = 0 \Rightarrow (x - a_i) \mid h, i = \overline{1, n}$$

$$(x - a_i, x - a_j) = 1, a_i \neq a_j$$

$$\Rightarrow \underbrace{(x - a_1)(x - a_2) \dots (x - a_{n+1})}_{\deg = n+1} \mid \underbrace{h(x)}_{\deg \leq n} \Rightarrow \underbrace{h(x)}_{\deg \leq n} = \underbrace{b(x)}_{\deg \leq n} \underbrace{(x - a_1)(x - a_2) \dots (x - a_{n+1})}_{\deg \geq n+1} \Rightarrow b(x) = 0 \Rightarrow f(x) = g(x)$$

